THE COMPOSITION OF TWO QUASI-AFFINE MORPHISMS IS QUASI-AFFINE

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 2280-2283 and proof span 2285-2314 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains an xymatrix fibre square; it is copied verbatim and the retained preamble loads Xy-pic.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The same-chapter characterization of quasi-affine morphisms by being a quasi-compact open subscheme of the relative spectrum of a quasi-coherent sheaf of algebras. Cited at proof lines 2506-2508. Represents the preimage under the second morphism inside a relative spectrum, which is the object that is then pushed forward.
- The pushforward of a quasi-coherent sheaf of algebras along a quasi-compact open immersion is quasi-coherent and pulls back to the original, cited from the schemes chapter. Cited at proof lines 2510-2513. Produces the sheaf of algebras on the affine scheme V whose relative spectrum makes the square a fibre square.
- The relative spectrum of a quasi-coherent sheaf of algebras, its compatibility with base change by an open immersion, and the fact that relative spectra over affine bases are affine. Cited at proof lines 2514-2520. Turns the open immersion j into a fibre square with an affine upper right corner, giving the affine scheme in which the double preimage sits as a quasi-compact open subscheme.

Source reference. Original lemma 01LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2102–2102. Same-chapter reference used in the retained author text at source line(s) 1780, 1839, 1841, 2273, 2311, 2337. It is supplied only as a navigation pointer so that the retained text compiles; the counted item

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Source reference. Original lemma 01SM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2343–2343. Same-chapter reference used in the retained author text at source line(s) 2290, 2323, 2338. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2081–2081. Same-chapter reference used in the retained author text at source line(s) 262, 404, 882, 1765, 1924, 2256, 2296, 8726, 9175, 10731, 13903. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *The composition of quasi-affine morphisms is quasi-affine.*

Proof. Let $f : X \to Y$ and $g : Y \to Z$ be quasi-affine morphisms. Let $U \subset Z$ be affine open. Then $g^{-1}(U)$ is quasi-affine by assumption on g. Let $j : g^{-1}(U) \to V$ be a quasi-compact open immersion into an affine scheme V. By Lemma 01SM above we see that $f^{-1}(g^{-1}(U))$ is a quasi-compact open subscheme of the relative spectrum $\underline{\mathrm{Spec}}_{g^{-1}(U)}(\mathcal{A})$ for some quasi-coherent sheaf of $\mathcal{O}_{g^{-1}(U)}$-algebras $\mathcal{A}$. By Schemes, Lemma 01LC the sheaf $\mathcal{A}' = j_*\mathcal{A}$ is a quasi-coherent sheaf of $\mathcal{O}_V$-algebras with the property that $j^*\mathcal{A}' = \mathcal{A}$. Hence we get a commutative diagram

$$\begin{array}{ccccc} f^{-1}(g^{-1}(U)) & \longrightarrow & \underline{\mathrm{Spec}}_{g^{-1}(U)}(\mathcal{A}) & \longrightarrow & \underline{\mathrm{Spec}}_V(\mathcal{A}') \\ & & \downarrow & & \downarrow \\ & & g^{-1}(U) & \xrightarrow{j} & V \end{array}$$

with the square being a fibre square, see Constructions, Lemma 01LX. Note that the upper right corner is an affine scheme. Hence $(g \circ f)^{-1}(U)$ is quasi-affine. □